Figure 7.9 An example of a possible situation where LRU eviction policy would purge most “best-selling” items from a cache, in favor of the more recently viewed ones which becomes popular during a peak of requests. In the figure, items are ordered according to their most recent view; the number of views is shown only to display how most recent views during a peak can wipe normal best-selling products: while LRU policy never guarantee that best-selling items would stay in the cache, the higher frequency with which they are accessed makes it more likely that they will stay in the cache (as it is likely they will go to the head of the queue before being purged). During a peak, if the number of items that suddenly become popular is large enough, they could fill the cache and force the usual items to be purged. This would anyway be a temporary side-effect, that will regress after the peak; in some situations, this could also be desirable, as items popular during a peak should be accessed faster than the regular best-selling ones: LRU has a more dynamic turnaround than LFU, as we will see.

7.5.1 So How Do We Choose?

Those are not the only possible strategies for cache purging. A web cache, for instance, could also consider the cost of retrieving some information, in terms of latency, compute time, and even actual costs when, like in our example, you need to pay an external service to retrieve some information.